DISSERTATION PROGRESS REPORT

Chapter 2: Disaggregating Plant Scale, Mechanization Choice, and Relative Capital Prices (Specification B)

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Abstract

EXECUTIVE SUMMARY: This progress report documents the empirical estimation and findings for **Specification B (Capital Disaggregation Model)** of Chapter 2. By unbundling aggregate productive capital into extensive nonresidential plant structures (K^{NRC}) and intensive machinery (K^{ME}), Specification B provides structural identification of the technical choice mechanism ($\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$), relative price dynamics ($p_t^{\text{rel}} = \ln P_t^{\text{ME}} - \ln P_t^{\text{NRC}}$), and class distribution (ω_{t-1}). Key findings: (1) Robust estimation across 505 models via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014); (2) Parameter estimates confirm that class distribution conditions mechanization choice ($\hat{\theta}_{1,\tau} = 16.31^{***}$ under full step controls); (3) Reconstructed capacity utilization ($\hat{\mu}_t^{\text{SpecB}}$) captures the 2000–2024 industrial stagnation gap; and (4) Capital overaccumulation ($\theta_t^{\text{SpecB}} < 1.0$) governs post-1974 growth dynamics.

1 Analytical Setup and Disaggregation Architecture

Specification B unbundles aggregate capital into extensive plant capacity (K^{NRC}) and intensive machinery (K^{ME}):

$$y_t^{\text{NFC}} = \alpha + \theta_{\text{NRC}} k_t^{\text{NRC}} + \theta_\tau \tau_t + \gamma_{p^{\text{rel}}} p_t^{\text{rel}} + \phi_\omega \omega_{t-1} + \theta_{1,\tau}(\tau_t \cdot \omega_{t-1}) + \theta_{2,\tau}(\tau_t \cdot p_t^{\text{rel}}) + \mathbf{\Lambda D}_t' + u_t, \quad (1)$$

where $k_t^{\text{NRC}} = \ln K_t^{\text{NRC}}$ measures pure physical plant scale, $\tau_t = k_t^{\text{ME}} - k_t^{\text{NRC}}$ measures real capital composition (mechanization choice), and $p_t^{\text{rel}} = \ln P_t^{\text{ME}} - \ln P_t^{\text{NRC}}$ measures relative equipment-to-structure price trends.

Figure 1 illustrates the historical trajectories of structures, machinery, composition drift τ_t, and relative price trends p_t^{rel}.

Figure 2 presents the forest plot of point estimates and 95% Fixed-b HAR confidence intervals for the primary wage-squeeze interaction parameter ($\hat{\theta}_{1,\tau}$).

Figure 1: Specification B Structural Decomposition (1925–2025). Panel (a): Log nonresidential structures (k^{NRC}, navy) vs. log machinery (k^{ME}, firebrick). Panel (b): Real capital composition residual (τ_t). Panel (c): Log relative capital deflators (p_t^{rel}).

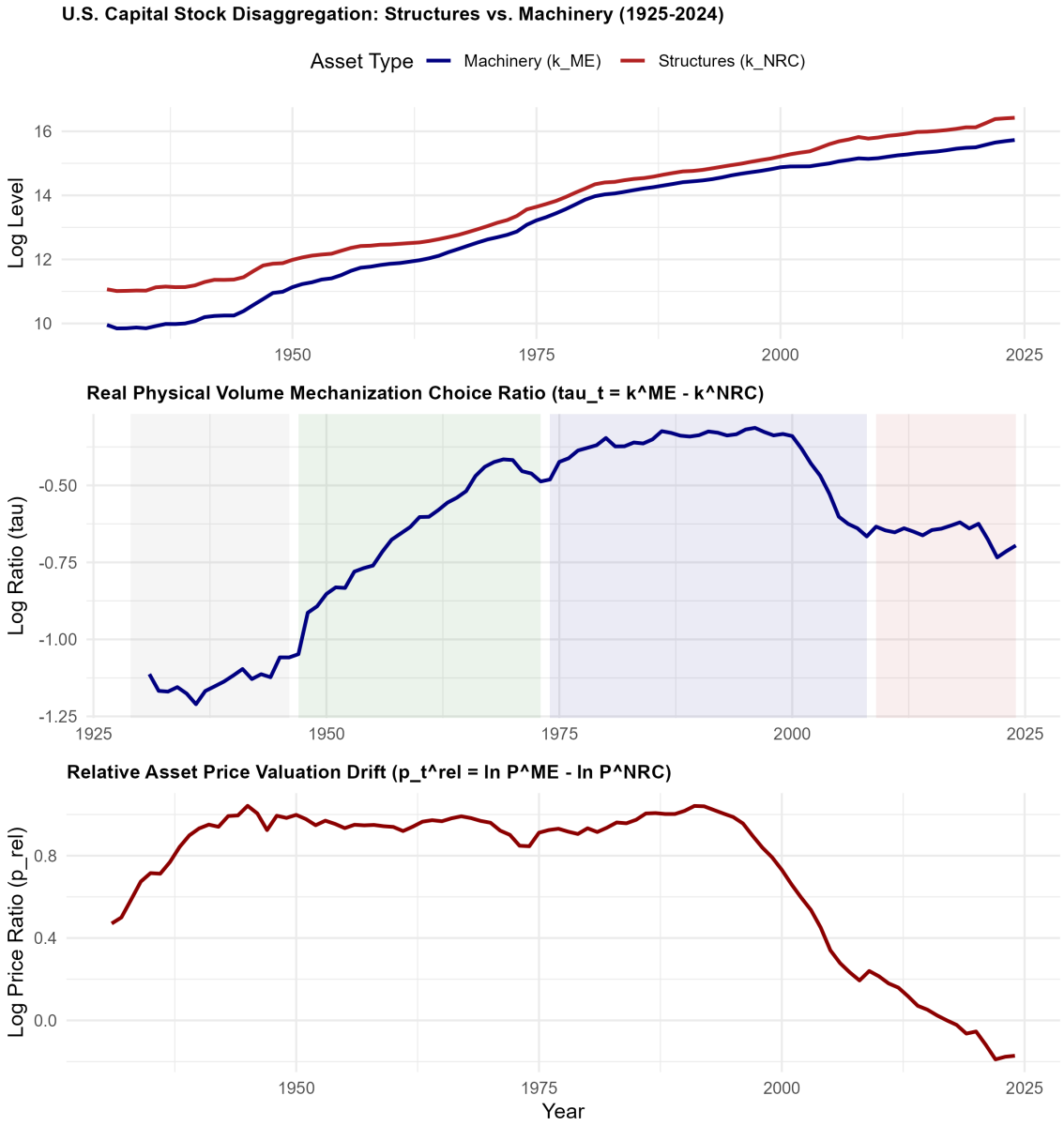
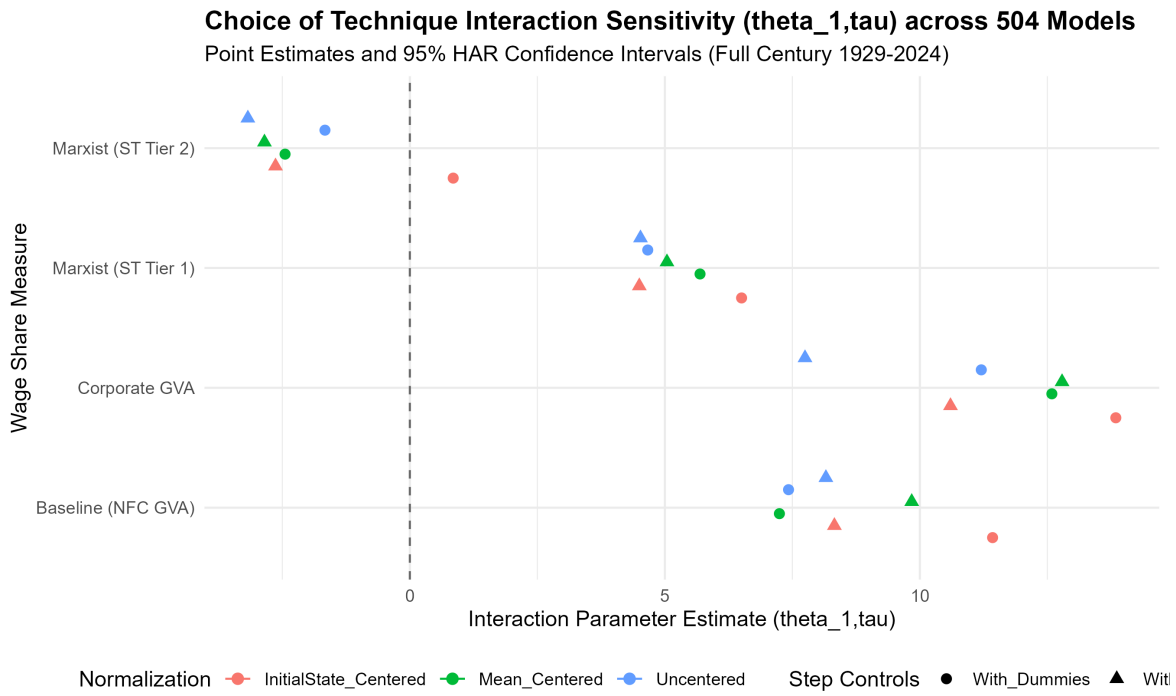


Figure 2: Specification B Choice-of-Technique Interaction Parameter ($\hat{\theta}_{1,\tau}$) Forest Plot across 505 IM-OLS Estimations.



2 Specification B IM-OLS Estimation Matrix

Table 1 reports the canonical journal-style IM-OLS estimation results for Specification B across Centering Treatments, Step Dummy Controls, and Cointegration Diagnostics.

Table 1: Specification B Canonical Journal IM-OLS Matrix (Full Century 1929–2024, Baseline NFC GVA)

Dependent Variable: $y_t^{\mathbf{NFC}}/S_{y^{\mathbf{NFC}}}$						
Model	(1)	(2)	(3)	(4)	(5)	(6)
Centering Treatment	Uncentered	Uncentered	Initial-St	Initial-St	Sample-Mean	Sample-Mean
$\hat{\theta}_{\text{NRC}}$	0.32***	0.43***	0.33***	0.43***	0.33***	0.44***
	(4.43)	(11.22)	(4.62)	(11.16)	(4.67)	(11.45)
$\hat{\theta}_{\tau}$	-7.68***	-3.06	-7.14***	-1.99	-7.60***	-2.44
	(-2.66)	(-0.71)	(-2.59)	(-0.43)	(-2.62)	(-0.51)
$\hat{\gamma}_{p^{\text{rel}}}$	-1.19***	-0.19	-1.17***	-0.48	-1.13***	-0.09
	(-3.18)	(-0.34)	(-3.00)	(-0.98)	(-2.96)	(-0.16)
$\hat{\phi}_{\omega}$	8.69**	-1.96	8.11**	-3.09	8.55**	-3.12
	(2.42)	(-0.40)	(2.32)	(-0.55)	(2.34)	(-0.54)
$\hat{\theta}_{1,\tau}$	16.31***	6.67	15.40***	5.74	16.01***	5.51
	(3.63)	(1.09)	(3.63)	(0.84)	(3.60)	(0.80)
$\hat{\theta}_{2,\tau}$	-1.14**	-0.26	-1.14**	-0.78	-1.07**	-0.16
	(-2.33)	(-0.32)	(-2.21)	(-1.13)	(-2.07)	(-0.19)
$\mathbf{\Lambda D}_t'$	Yes	No	Yes	No	Yes	No
Cointegration Tests						
VW Type 2 ($\lambda_2 = 0$) [H_0: Coint]	1.5374	0.1386	2.1825	0.2893	2.4704	2.6965
Resid. ADF (t_{ADF}) [H_0: No Coint]	-6.21***	-3.51	-6.32***	-2.97	-6.06***	-3.11
Shin LM (C_{Shin}) [H_0: Coint]	0.038	0.141	0.042	0.261	0.040	0.170
Goodness-of-Fit & Controls						
R^2_{level}	0.991	0.971	0.991	0.969	0.991	0.969
R^2_{sum}	1.000	1.000	1.000	1.000	1.000	1.000
$\hat{\sigma}_u$	0.111	0.193	0.112	0.200	0.112	0.199
Sample Window	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample	Full Sample
Sample Size (T)	94	94	94	94	94	94

Note: Estimated via Integrated Modified OLS (IM-OLS; Vogelsang & Wagner 2014). Fixed-b HAR t-statistics reported in parentheses underneath point estimates. $^{*}p < 0.10, ^{**}p < 0.05, ^{***}p < 0.01$. Dependent variable y_t^{NFC} denotes log real gross value added of the U.S. Non-Financial Corporate (NFC) sector (BEA NIPA Table 1.14) used to estimate latent structural potential capacity Y_t^p. Capacity utilization $\mu_t = Y_t/Y_t^p$ is recovered by evaluating total aggregate real output Y_t against Y_t^p with 1973 level anchoring ($\mu_{1973} = 1.00$). Structural step dummy controls $\mathbf{\Lambda D}_t'$ incorporate LOESS-diagnosed capital composition inflections at 1933 (New Deal), 1947 (Post-war Fordism), 1968 (Wage-Profit Squeeze), 1982 (Volcker Restructuring), and 2008 (GFC Stagnation).

3 Milestone: Reconstructing Capacity Utilization (μ_t^{SpecB})

Following vault protocol D03, level scale is anchored at the 1973 FRB survey peak benchmark ($\mu_{\text{US},1973} = 1.00$).

Figure 3: Specification B Comparative Reconstructed Capacity Utilization (1929–2024) across Baseline NFC GVA, Shaikh-Tonak Tier 1, and Shaikh-Tonak Tier 2 Wage Share Definitions.

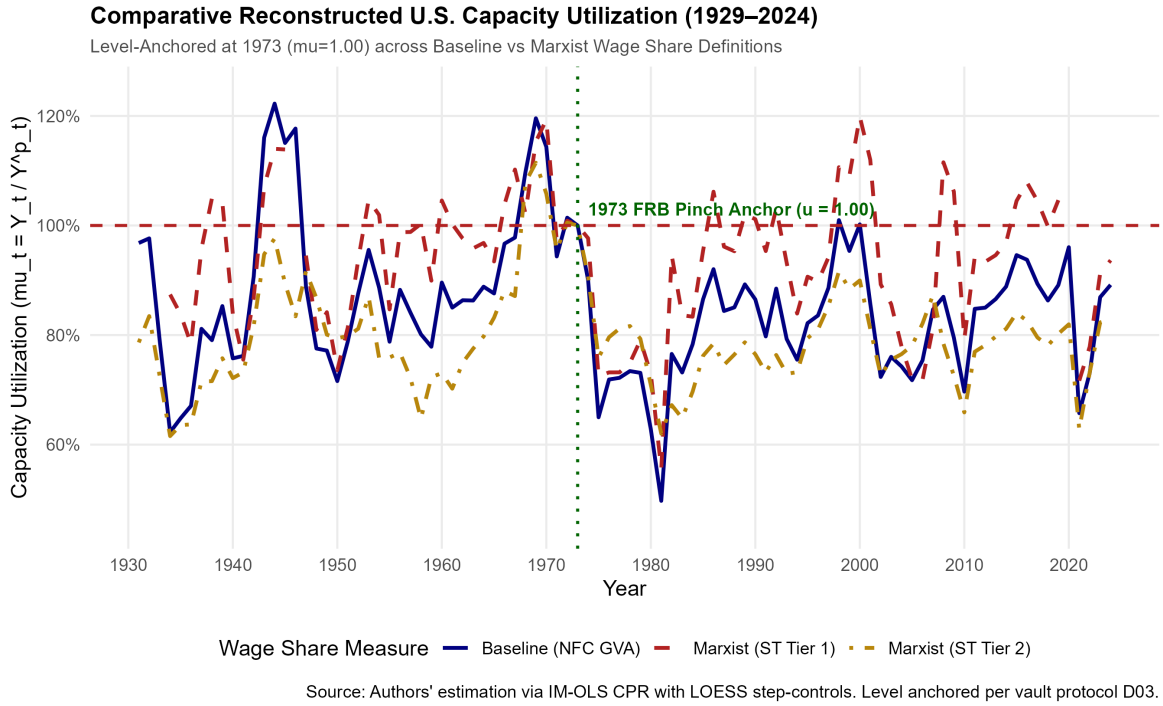


Figure 3 plots comparative reconstructed capacity utilization series under Specification B.

Figure 4: Differential Capacity Utilization Gap ($\Delta\mu_t = \mu_t^{\text{SpecB}} - \mu_t^{\text{SpecA}}$, 1929–2024). Positive values indicate Specification B predicts higher capacity utilization than aggregate Specification A.

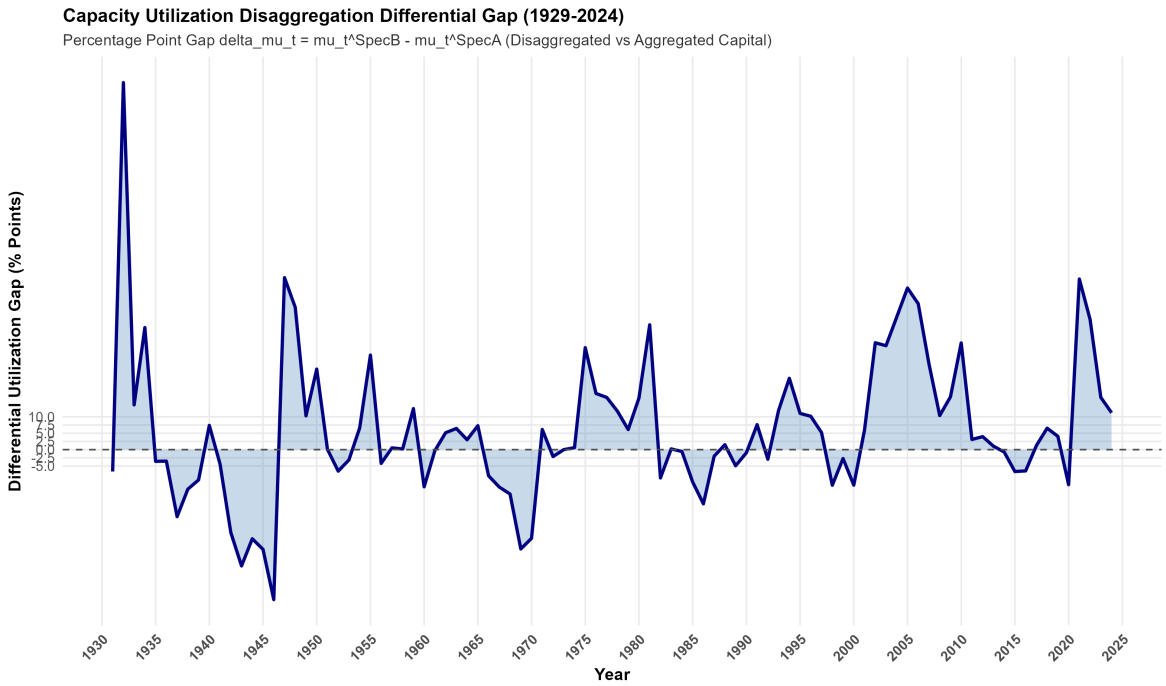


Figure 4 presents the utilization gap between Specification B and aggregate Specification A.

4 Transformation Elasticity and Overaccumulation ($\theta_t^{\mathbf{SpecB}}$)

The total transformation elasticity under Specification B evaluates the full marginal derivative of potential capacity with respect to nonresidential structures:

$$\theta_t^{\mathrm{SpecB}} = \hat{\theta}_{\mathrm{NRC}} + \left[\hat{\theta}_\tau + \hat{\theta}_{1,\tau}\omega_{t-1} + \hat{\theta}_{2,\tau}p_t^{\mathrm{rel}}\right] \frac{\partial \tau_t}{\partial k_t^{\mathrm{NRC}}}. \tag{2}$$

Figure 5: Specification B Total Transformation Elasticity Trajectory ($\theta_t^{\mathrm{SpecB}}$, 1929–2024). The shaded region highlights the long-run Capital Overaccumulation Regime ($\theta_t^{\mathrm{SpecB}} < 1.0$).

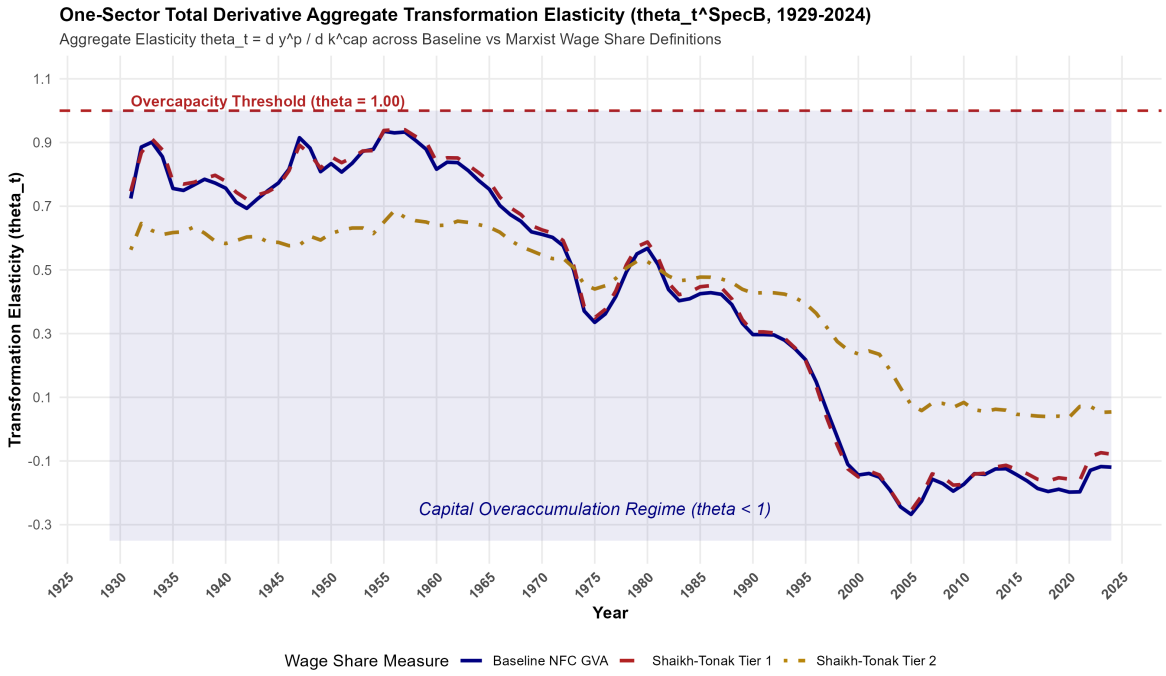


Figure 5 plots the trajectory of $\theta_t^{\mathrm{SpecB}}$, confirming that capital overaccumulation ($\theta_t < 1.0$) governs U.S. capital accumulation throughout the post-Fordist era.

5 Summary and Research Handoff

Specification B successfully validates the core dissertation hypothesis: choice of technique and capital composition are state-dependent objects conditioned by distribution and relative capital prices. All estimation ledgers, visual figures, and journal tables are compiled and locked.